

Uday Bhaskar Kale

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Eligible for student employment (Werkstudent, up to 20 hrs/week)

Education

2024 - Oct 2026 (expected)	M.Sc. Cognitive Systems University of Potsdam, Germany Focus: Natural Language Processing, Machine Learning, Computational Linguistics, Cognitive Architectures, Agentic-AI
2017 - 2021	B.E. Computer Engineering University of Pune, India CGPA: 8.93 / 10

Professional Experience

Jan 2023 - May 2024	Technical Lead / Co-Founder, Seerlink Analytics Pvt Ltd, India - Built and deployed two Flutter/Firebase applications supporting dairy supply-chain management. - Designed data validation and reporting pipelines for daily milk collection records. - Led development from requirements gathering through deployment and maintenance.
Aug 2021 - Dec 2022	Application Development Analyst, Accenture, India - Developed and maintained SAP PI/PO integrations across enterprise production systems. - Implemented data mapping and interface orchestration for reliable data exchange. - Supported deployment, debugging, and production issue resolution. - Collaborated with cross-functional teams to improve system reliability.

Research & Technical Projects

Kakapo - Corpus Query and Annotation Platform (Master's Thesis)	<i>Python, FastAPI, SQLite, Docker, ChromaDB, spaCy</i> - Designed and developed a corpus management platform for computational linguistics research. - Built automated pipelines to transform hierarchical corpora into normalized SQLite databases. - Implemented annotation workflows, role-based access control, SQL querying, backup/restore, and administrative tools. - Currently integrating Retrieval-Augmented Generation and Natural Language to SQL interfaces for interactive corpus exploration. - Planned deployment within the University of Potsdam research environment.
Argument-Aware Retrieval for Automated Fact Checking (Research Module)	<i>Python, SentenceTransformers, BM25+Dense, Transformers, Qwen, Cerebras</i> - Developed an argument-aware retrieval pipeline for evidence retrieval using the LIARArg benchmark. - Combined hybrid retrieval, NLI-based reranking, and probabilistic reasoning with language models. - Evaluated against flat retrieval baselines using reproducible experimental pipelines. - Improved fact-checking performance through argument-structure-aware retrieval.
Geo AI Agent (Arcadis Hackathon Prototype)	<i>Python, GeoPandas, Shapely, Ollama, Streamlit</i> Built a multi agent system for geospatial site risk assessment.

	• Combined GIS data, planning documents, and local LLMs to generate evidence based reports with interactive maps.
Turix (Python Library)	<i>Python, FastAPI</i> • Designed an execution authority layer for AI systems that authorizes actions before execution. • Implemented policy evaluation, audit logging, fail safe execution, and tenant aware authorization.
Commit Message-Based Bug Localization	<i>Python, scikit-learn, LangGraph, Groq</i> • Built a multi-label classifier predicting affected source files from commit messages. • Integrated LangGraph for semantic reranking of ML predictions. • Evaluated using Precision@K on real-world software repositories.

Technical Skills

Programming	Python, Java, SQL, NoSQL
Machine Learning	PyTorch, Hugging Face Transformers, scikit learn, LangGraph
NLP & LLMs	Natural Language Processing, RAG, AI Agents, Prompt Engineering, Model Evaluation
Data Analysis	pandas, NumPy, FAISS, Data Pipelines, Feature Engineering
Software Engineering	FastAPI, Streamlit, Git, Docker, REST APIs
Enterprise	SAP PI/PO, SAP Basis

Languages

English - Professional, German - A2 (currently learning), Hindi - Native, Marathi - Native